



Graphs in Machine Learning

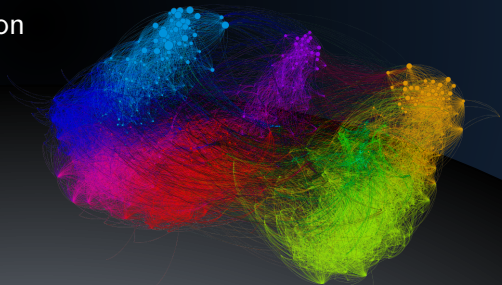
Introduction to Graphs in ML

Course Overview and Motivation

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Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns



What will you learn in the Graphs in ML course?

Concepts

What will you learn in the Graphs in ML course?

Concepts, tools

What will you learn in the Graphs in ML course?

Concepts, tools, and methods

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One example: Online Semi-Supervised Face Recognition

Two (main) sources of graphs in ML

Natural graphs as models for networks

Constructed graphs as nonparametric basis

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Constructed graphs as nonparametric basis

- we create (learn) the similarity structure from flat data
- it's a tool (e.g., nonparametric regularizer) to encode structural properties (e.g., independence, ...)

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<https://misovalko.github.io/mva-ml-graphs.html>